GUIDELINES: Impact of Strategies on Data Center Energy Consumption, Carbon Footprint, and Water Consumption

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Executive Summary

This article summarizes the interplay between water, energy, and carbon in data center design and operation strategies. The manuscript first describes equipment and facility cooling strategies, and then discusses how the interplay is exhibited when comparing key data center performance and environmental metrics, culminating in a summary table describing the interrelationships. Key elements of this work include (1) the relationship between cooling system energy efficiency and water consumption, (2) the role of geographical location on energy efficiency, water consumption, and environmental burden; and (3) a holistic analysis of water consumption, carbon footprint, and water scarcity footprint should be considered concurrently. The use of waste heat is examined for reducing the holistic environmental burden of the data center and surrounding buildings. A metrics calculation scheme with examples is provided with the intention of industry adoption for aiding in site planning for new data centers.

Date: March, 2026

Table of Contents

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1. Overview of Cooling Strategies

Data center cooling systems transfer the heat generated by information technology equipment (ITE) to an outside heatsink. This heatsink is typically the ambient air outside of the data center but can also be for heat recovery (e.g., a district heating network or an industrial process such as pre-heating feed to a wastewater digester). The relationship between energy, water, carbon, water scarcity, and heat reuse in these cooling systems depends heavily on the chosen process for heat transfer to the heatsink. This paper therefore provides a high-level categorization of different cooling system approaches and includes illustrative examples selected to highlight various aspects of the interplay between these metrics. The goals of the paper are to provide guidelines on questions to consider when determining the environmental burden associated with data center cooling system design decisions.

1.1 COOLING THE ITE

Two primary means for cooling the ITE directly are (1) traditional air-based cooling, and (2) liquid cooling.

1.1.1 Air-Based ITE Cooling

The vast majority of data centers as of 2024 still employ air-cooled servers, and air-cooling will, most likely, be around for the foreseeable future. Technologies involving the use of air directly removing heat from ITE are generally mature. The most common approach is to use a finned heatsink to minimize the thermal resistance between heat sources and the airstream. The heatsink can include a dedicated fan, and the server chassis can include fans to pull air through the server. Other technologies related to air-based cooling include the use of heat spreaders or heat pipes to distribute the heat in a manner to facilitate efficient cooling.

As air passes over the ITE it absorbs heat, cooling the equipment. The heated air is then either transferred to a water or refrigerant cooling loop or exhausted from the room. The chilled water or cool supply air is often provided as the output of a refrigeration cycle in the form of a Computer Room Air Conditioner (CRAC) for air-refrigerant heat exchange, or the combination of a Computer Room Handler (CRAH) for air-chilled water heat exchange with heat removed downstream from the chilled water loop by a chiller unit.

1.1.2 Liquid ITE Cooling

Liquid cooling is gaining market share for a number of reasons, notably the densification of ITE associated with large language model (LLM) training. Recent and proposed builds for this purpose have surpassed the gigawatt scale for data center campuses, many of which contain racks in excess of 100 kW and therefore require liquid cooling to expunge the vast amount of ITE heat output. This industry driver therefore has made liquid cooling increasingly relevant and could potentially dominate the ITE cooling market.

Liquid cooling technologies can generally be categorized in two ways:

1 - Direct liquid cooling (DLC) or cold plate, direct to chip (single-phase and two-phase)

2 - Immersion Cooling (single phase and two phase)

Figure 1 indicates the general cooling system configurations for both of these cooling strategies. In liquid cooling systems, the heat generated by ITE is transferred to a liquid fluid either through direct contact or through a conductive plate. Liquid media such as water or refrigerant can hold a much larger amount of heat per unit volume than air, so heat can therefore be removed from ITE much more efficiently. Once the heat is transferred to the liquid medium, it can then be removed from the room either through a separate chilled water loop or exhausted into an airstream.

Liquid cooling systems may allow the technology cooling system (TCS) loop immersion fluid to operate at elevated temperatures. For a chilled water system for the facility cooling, this would allow the chilled water loop temperatures to increase from the temperatures typical in air-cooled facilities. As these temperatures increase the requirements for mechanical cooling or evaporative cooling towers can be eliminated in some locations.

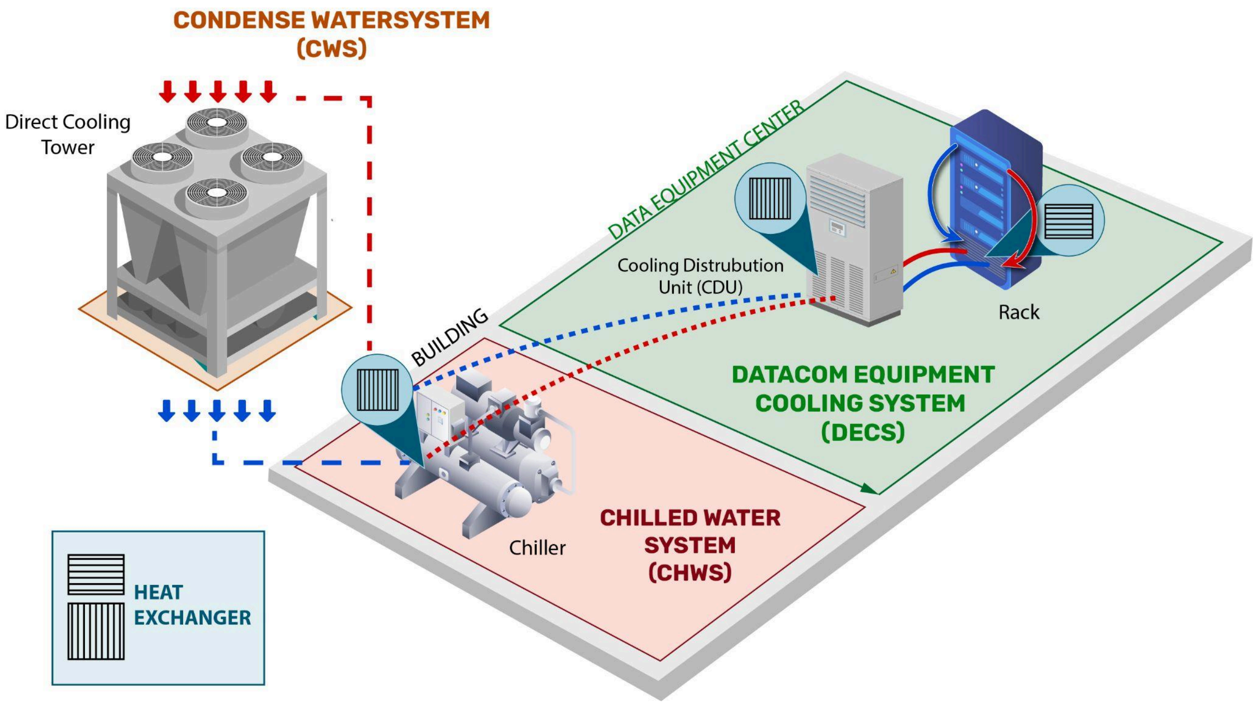

Figure 1. Typical DLC liquid cooling system with compression cooling

The use of chillerless and dry cooling, naturally, is conditional to the temperatures the chip manufacturers suggest for their equipment and the location of the facility. If the IT manufacturer requires a maximum operating temperature of 32 degrees C, for example, then there could be occasions when the use of compression cooling (i.e., a CRAC or chiller) will be necessary. Naturally, this is even more true if the required temperature is even lower (e.g., 17 degrees C).

A liquid cooling technology, with certain exceptions such as extreme high end IT equipment (e.g., artificial intelligence- AI) increases the possibility that the data center operator could run the cooling systems year around without compression cooling.

These options provide the opportunity to run the TCS, which is the part of the cooling system that directly cools the IT equipment, at higher temperatures. This capability is desirable because running cooling systems at 40-50 degrees C enables the use of dry air coolers for most of the year, almost everywhere in the world, without the need to use compression cooling. Eliminating compression cooling greatly reduces the cooling system energy consumption, thereby making the cooling system more energy efficient.

The currently most common liquid cooling approach is DLC mainly because it can be deployed with minimal changes in the server room (provided that a water loop is present).

1.1.3 Other Cooling Strategies

Other cooling strategies involve the use of air to remove heat directly from the ITE, but the cooling of the airstream occurs closer to the ITE than in a CRAC, CRAH, or direct air cooling system typical of legacy air-cooled data centers. These cooling strategies are sometimes referred to as a form of liquid cooling or hybrid liquid-air cooling at a rack level:

- Rear door cooling involves the use of cooling of exhaust air from the servers through the rear of the rack through a built-in refrigerant or water heat exchanger.
- In-row cooling contains enclosed air streams within cabinets and heat exchange with a cooling coil inside the row of racks.
- Overhead cooling contains units above the racks, drawing in air from the hot aisle, removing heat via a cooling coil, and exhausting the cooled air into the cold aisle.

1.2 FACILITY COOLING

The three main media for transferring heat from the data halls to an external location are refrigerant, water, and air. Refrigerant-based systems such as Direct Expansion (DX) or CRAC units contain an evaporator unit in the data hall. Refrigeration pipework is run from the evaporator unit to the condenser unit in a remote location as shown in Figure 2. This arrangement is similar to cooling systems found in residential and commercial applications. These types of systems are usually non-evaporative and consume no onsite water. Refrigerant-based systems are typically modular, making them good options for smaller facilities or facilities requiring the flexibility to easily scale their cooling needs. However, these systems are usually less efficient and consume more electricity on a per-ton of cooling basis compared to larger chiller systems.

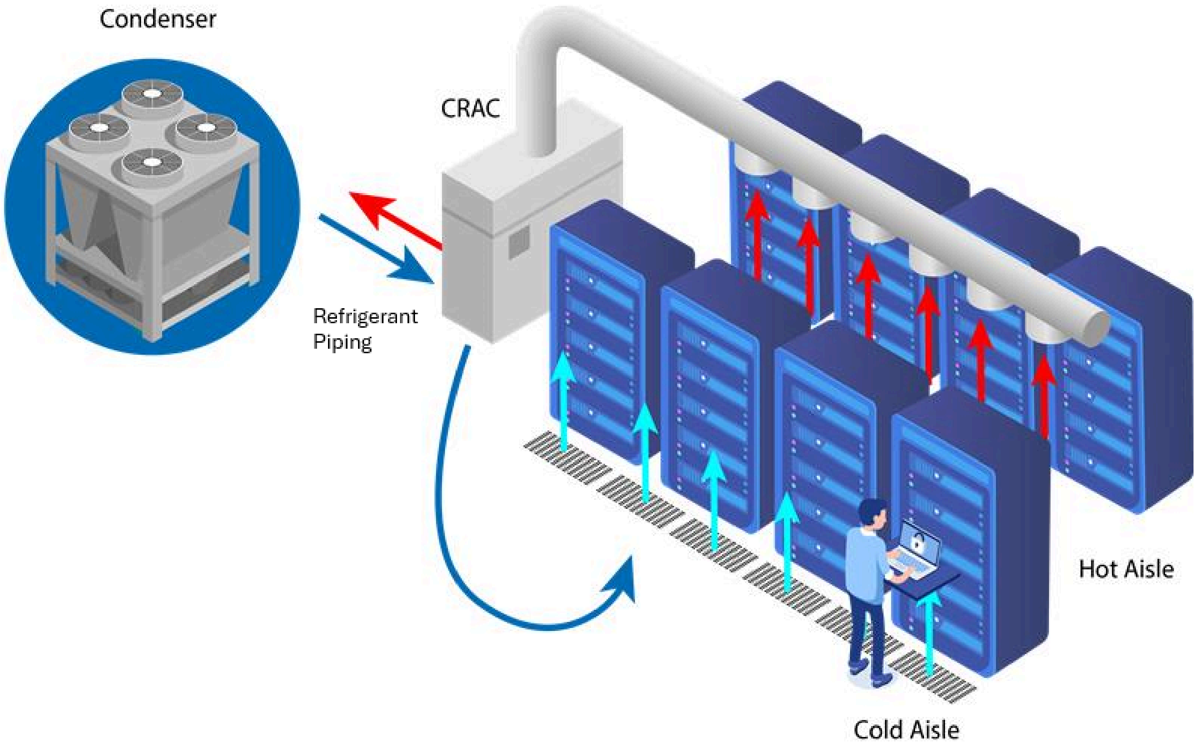

Figure 2. Refrigerant-based system containing a CRAC unit with external condenser

Water-based systems contain a chilled water loop to collect heat from the IT equipment through cooling equipment such as a CRAH, a rear door heat exchanger, or even liquid cooling systems such as direct-to-chip liquid cooling. Figure 3 shows a CRAH-based system. The heated water in the chilled water loop is sent to a heat rejection system such as a chiller (as shown in Figure 3) or directly to a cooling tower. Chillers can either be air-cooled or water-cooled, and cooling towers can either be evaporative or non-evaporative. Systems with a chilled water loop require more physical infrastructure compared to refrigerant-based systems, but water-based systems allow for increased flexibility with the use of IT equipment in the facility. Once a chilled water system is plumbed into a data hall, that same infrastructure can be used for both air cooling of equipment and liquid cooling with limited retrofit work. This type of system also allows for a uniform design in the data hall, with flexibility in the use of heat rejection equipment. A single chilled water data hall design could be used with options for air-cooled chillers, water-cooled chillers, or even a chiller-less ("warm water") system using only cooling towers under allowable local climate conditions. This operational flexibility may be critical to some data center operators due to increasing ITE power density.

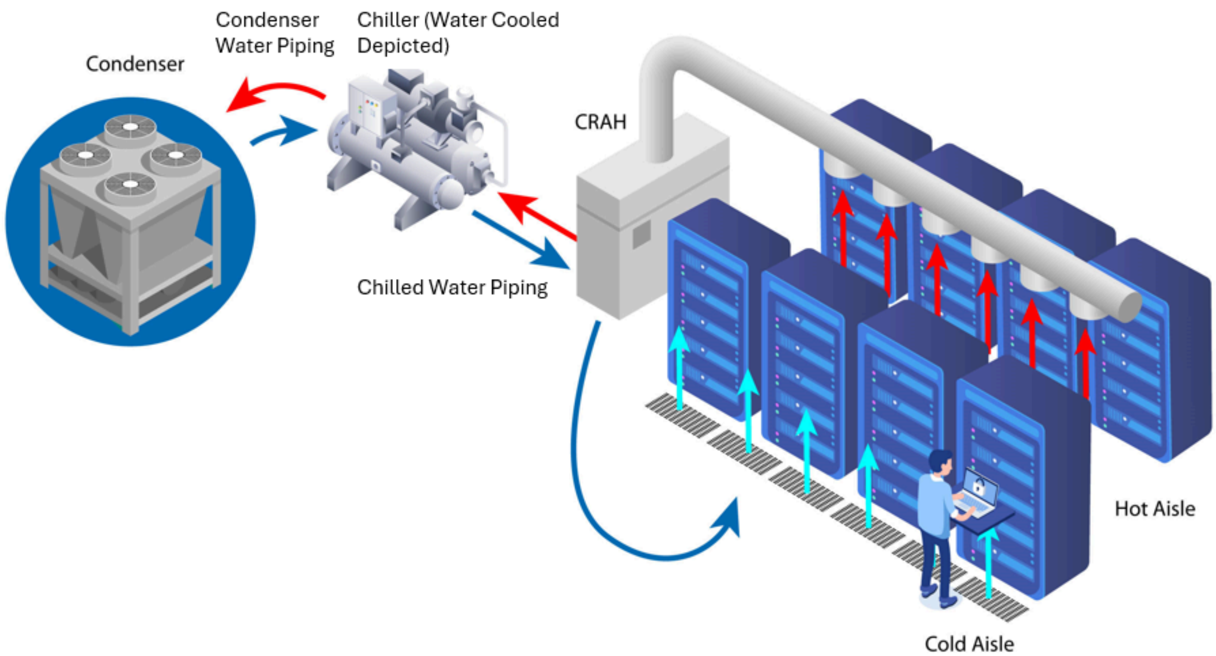

Figure 3. Chilled water-based system with a chilled water loop between a chiller and CRAH unit, subfloor air supply and open air return to the CRAH unit

Air-based systems (Figure 4) use air as a means of removing heat from the data hall. In direct air-cooling systems, also known as free-air cooling or airside economization, outside air is brought directly into the data hall to absorb heat before being exhausted from the facility. Cooling and humidification of the supply airstream can be achieved with an adiabatic system such as cooling media or a water spray system. In an indirect air-cooling system, outside air uses a heat exchanger to remove heat from an internal circulating airstream. An indirect system can be used to cool the air inside the data hall without the increased humidity associated with direct air systems. Indirect air cooling systems have a lower efficiency than direct fresh air systems due to heat transfer losses through the heat exchanger, but indirect air cooling is useful in situations with poor outdoor air quality or in humid locations.

Purpose-built chiller-less air-cooled data centers can have very low power consumption since ventilation fans use less electricity compared to compressor-based cooling systems. However, as the power density of IT equipment increases, the limits of air cooling for certain computing applications may be reached. As the power density of IT equipment increases, the ability of air to cool equipment becomes limited, thereby necessitating increased airflow, which results in increased power consumption by ventilation fans.

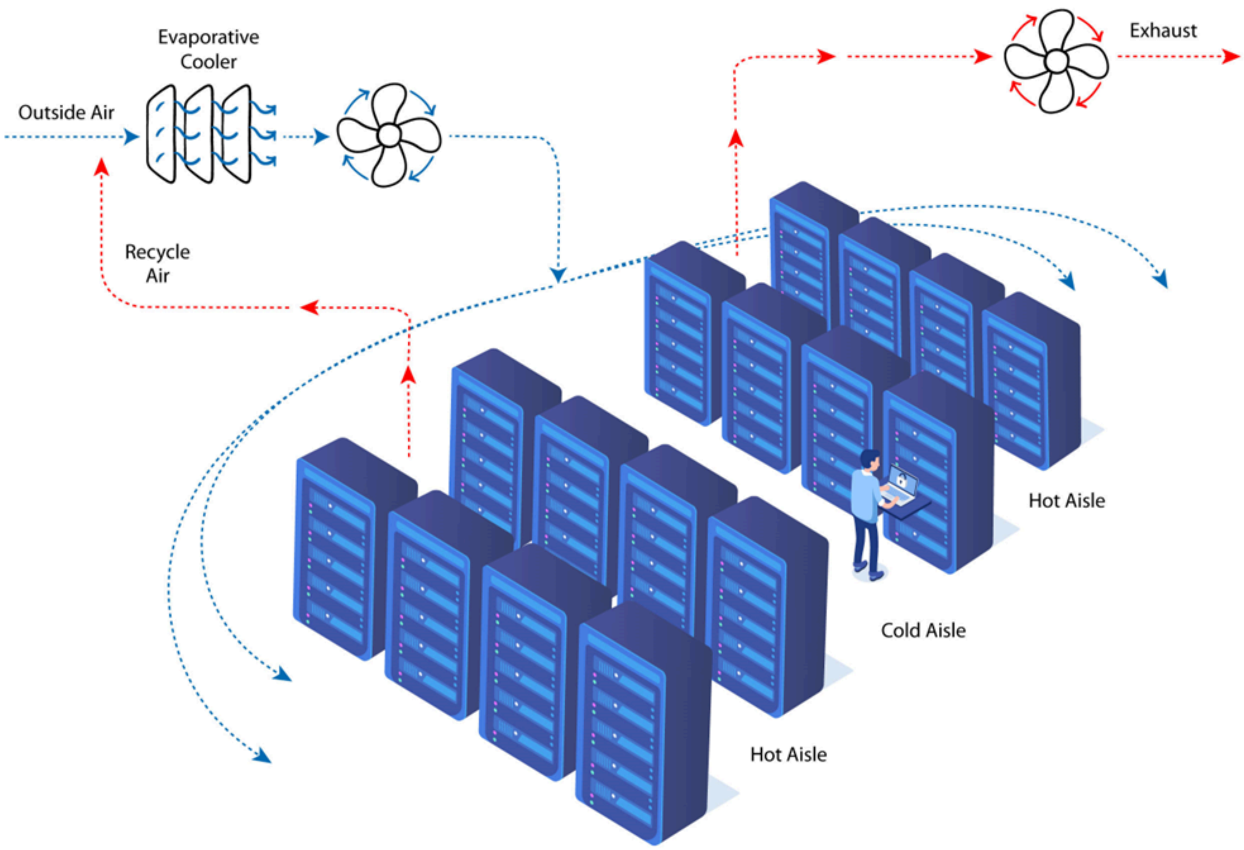

Figure 4. Direct air-based system featuring an evaporative cooler and ducted supply and return. Blue and red arrows indicate cold and hot airflows, respectively.

1.3 COOLING SYSTEM COSTS AND PLANNING

Each ITE and facility cooling strategy has its own independent capital and operating costs, and further analysis is required to compare the total cost of ownership (TCO) between different systems. Liquid cooled systems, for example, may have higher capital costs or specialized maintenance needs that could be barriers for some operators. However, the reported lower PUE values associated with liquid cooling systems and the higher waste heat recovery potential (due to higher waste heat temperatures) likely translates to lower power consumption requirements. Since power is a larger utility cost than water for data centers, the relative lower power consumption for liquid cooled systems in turn means a lower operating cost when not accounting for maintenance costs.

Another element of cost analysis involves planning for future scenarios. Two potential trends are related to water and power trends. Regarding water, there could be increased water scarcity due to prolonged droughts such that the local water draws may not be met, and as a result the cost of water can increase significantly. While weather is uncertain, some evidence of the reduction in water tables in specific areas has been seen [1]. Regarding carbon, the use of carbon credits or taxation could add significant costs if the location falls under a political region that enacts this policy. Canada, for example, has the Greenhouse Gas Pollution Pricing Act (GHGPPA) carbon tax [2], whereas South Korea, and the European Union have the Korea Emissions Trading Scheme (K-ETS) [3] and European Union Emissions Trading System (EUETS) [4] cap and trade programs, respectively.

2. Interplay Between Energy, Water, and Carbon

The data center industry is booming, which will result in the continuous development and planning of new data centers. It is important to understand the interplay between data center energy consumption, water consumption, carbon footprint, and water scarcity footprint. It is furthermore essential to reuse waste heat to reduce energy consumption in cooling systems, which in turn reduces the carbon and water scarcity footprints associated with facility operation.

Data center cooling systems generally consume energy and water, and they therefore have carbon and water scarcity footprints. There is furthermore the relationship between cooling system design and waste heat recovery potential. The general relationships between these areas are listed in the following sections.

2.1 RELATIONSHIP BETWEEN ENERGY AND WATER

Many cooling systems leverage the evaporation of water to aid in the cooling process either onsite or at the source of power generation. Onsite water consumption is typically due to evaporative cooling systems such as cooling towers or evaporative coolers, while offsite water consumption is due to the consumption of water at the source of power generation. Both onsite and offsite water consumption should be considered as part of a holistic review and optimization of resource consumption.

2.1.1 Onsite Water Consumption

Section 1.2 provides a general indication of onsite water consumption by cooling systems. It is useful to compare systems that use the evaporation of water and systems that do not use the evaporation of water to support cooling operation. Table 1 provides lists of cooling systems that do and do not contain evaporation. Systems that do not directly evaporate water into the environment generally use an air-to-liquid heat exchanger like an air-cooled condenser to complete the final heat transfer in the system either directly from a chilled water loop or to reject heat from a refrigeration-based system. In many climates, non-evaporative cooling systems use more electricity compared to systems that evaporate water since non-evaporative cooling systems can only reach the dry bulb temperature of the air.

Systems that evaporate water use the latent heat of evaporation to decrease the temperature of cooling water or air stream. In a typical cooling tower, evaporating just 1% of the water stream decreases the water temperature by around 10 F. Cooling water or air can be decreased to a value approaching the web-bulb temperature of the ambient air, which in hot dry climates can be more than 30 degrees F lower than the dry bulb temperature during peak cooling season. The additional temperature decrease allows water-cooled chillers to run more efficiently by cooling the refrigerant in the condenser to a lower temperature, thus decreasing the amount of compressor work. However, the decrease in electricity usage comes at the cost of onsite water consumption.

Table 1. Examples of Evaporative and Non-Evaporative Cooling Systems

Evaporative	Non-Evaporative
Direct and indirect evaporative cooler	Direct expansion (DX)
Water-cooled chiller with wet cooling tower	Water-cooled chiller with dry cooling tower
Chiller-less chilled water system with wet cooling tower	Air-cooled chiller

The following electricity and water use trends can be applied to evaporative and non-evaporative chiller-based systems when economization is not used:

- Non-evaporative systems will see an increase in electricity usage during the summer when the dry bulb temperature is at peak levels. In refrigerant-based systems, the increased dry bulb temperature will result in a higher temperature of the refrigerant leaving the condenser, requiring the compressor to work harder to meet the same conditions in the evaporator.
- Evaporative systems will generally see a more consistent rate of electricity consumption throughout the year. However, evaporative systems consume more water as the wet bulb temperature increases.

Cooling systems are designed to provide a certain capacity at worst-case conditions, though for the majority of operational hours, the system will operate at partial capacity and/or with some level of economization. While the design day values are important for the sizing of the system and ancillary components, simply comparing full load conditions provides an inaccurate assessment of power and water consumption over the varying ambient and heat load conditions the facility encounters throughout annual operating conditions. It is important to compare systems with metrics that account for the varying operational conditions, as partial load conditions are frequently less efficient and some locations will have the ability to operate in economization mode a large portion of the year. The following section reviews various options for system economization.

The level of cooling efficiency, expressed as kW of cooling per gallon of water consumed, depends highly on the evaporative cooling system type (e.g., humidification of ventilation air, open loop cooling tower, closed loop with adiabatic assist) and local climate. The reader is referred to existing academic studies that compare different cooling system configurations and their impact on energy efficiency and water usage (e.g., [5]).

2.1.2 Economization of Site Cooling Systems

An “economizer” is not a piece of equipment or an object but rather a mode of operation that allows a resource-intensive part of the cooling system, such as a chiller, to shut down or reduce load if ambient conditions allow. Economization can be performed in three different manners: air-side, water-side, or refrigerant-based.

Air-side: Outdoor air is brought directly into the data hall when the temperature and humidity are within acceptable ranges. It is possible to adjust the temperature and humidity with the use of evaporative cooling systems. Alternatively, outdoor air can be used to cool indoor air through a heat exchanger. The latter method, often called indirect air cooling, is less efficient due to heat loss through the heat exchanger but does not introduce outside humidity and contaminates into the data hall.

Air-side economization can be utilized as the primary means of cooling in many locations when it is used with evaporative cooling systems.

It is important to note that indirect air cooling provides the option to utilize evaporative cooling without the need to humidify air into the facility. Indirect air cooling features two airstreams: an outside air stream that could be cooled adiabatically using evaporative cooling, thereby increasing the humidity ratio while reducing the wet bulb temperature; and a second stream of supply air that can be a combination of recirculated internal air and ventilation air. In indirect air cooling, the outside air stream’s dry bulb temperature is below that for the supply air, and heat is exchanged between the two streams via a heat exchanger. The first, potentially humidified stream is exhausted to the outside environment, resulting in a cooled supply airstream.

Water-side: Chilled water from the cooling system either fully or partially bypasses the mechanical cooling equipment and rejects heat directly to equipment such as a fluid cooler or cooling tower. Alternatively, water-side economization can include pre-cooling chilled water return prior to entering the chiller.

There are many possible water-side economization methods. For example, water-cooled chillers can be operated in an economizer mode by using an additional heat exchanger to cool the chilled water system with the condenser water when ambient conditions allow. A heat exchanger is used to separate the chilled water system from the condenser loop as the latter is typically maintained at a lower water quality. The placement of the heat exchanger either in series or parallel with the chiller will determine if the system can be operated in a partial economization mode or if the system must be operated with either mechanical cooling or in full economization mode. When the heat exchanger is in series with the chiller on the chilled water loop, the heat exchanger can pre-cool the chilled water return and reduce the mechanical load on the chiller when conditions do not allow for complete cooling with the condenser water. For systems that use evaporative cooling towers, the evaporation of water will cool the water to within a couple of degrees of the ambient wet bulb temperature. Therefore, the economization potential depends on wet bulb temperatures. Systems that use dry cooling towers will be dependent on the dry bulb temperature as they do not utilize the evaporation of water for cooling. The wet bulb temperature can be 30 degrees F lower than the dry bulb temperature in hot and dry climates.

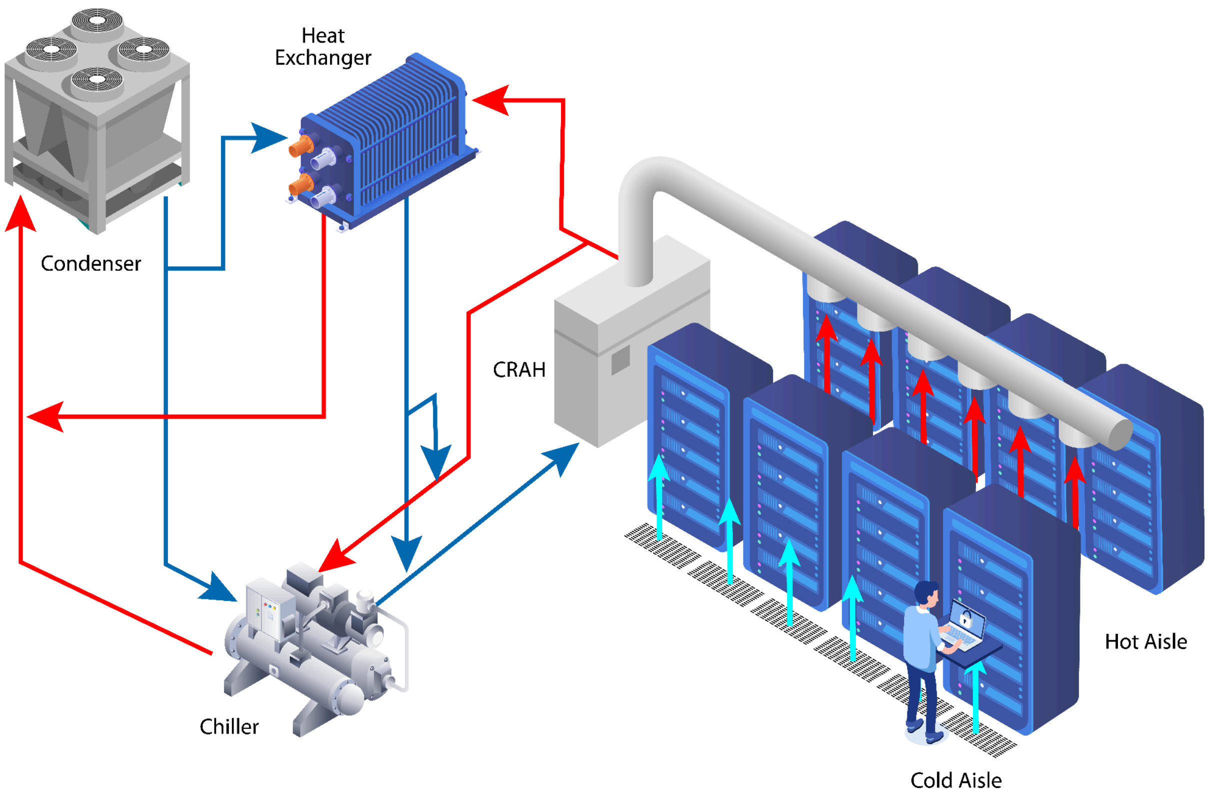

Figure 5. Water-cooled chiller with water-based economization

Air-cooled chillers can be operated in an economization mode in a similar manner to water-cooled chillers by adding a dry cooling tower to the system. Chilled water is fully or partially cooled through the dry cooling tower, allowing for a partial or full reduction in chiller operation. While not a requirement, this type of system is typically integrated directly into the air-cooled chiller system.

For both water-cooled and air-cooled chillers with dry cooling towers, the cooling potential of the dry cooler tower can be increased by adding pre-cooling by either a water spray or wetted media (i.e., adiabatic cooling) for the airstream. This type of system can allow the dry cooling tower to achieve temperatures lower than the dry bulb of air and likely offset power consumption and decrease the equipment size though additional water consumption will be required.

Refrigerant-based: Refrigerant in a system such as a direct expansion (DX) system is used to complete the refrigeration cycle without the use of mechanical compression. Refrigerant-based economizer systems are proprietary designs that depend heavily on the specific system design and the type of refrigerant used. Typically, these systems are able to switch from the use of an energy-intensive compressor to a pump to move the refrigerant when ambient conditions allow.

Cooling systems can either be economized with an air-side, water-side, or refrigerant-based system or a combination of more than one though this is generally not seen in industry due to infrastructure costs.

The ability to use economization at a facility is dependent on the temperature and humidity envelope in which the facility operates and the ambient weather conditions throughout the year. Many locations in the United States can operate in economization for the majority of the year if the facility utilizes the ASHRAE TC9.9 Thermal Guidelines allowable temperature and humidity envelopes. The Green Grid has put together the maps shown in Figures 6 through 8.

The potential for economization increases as the temperature inside the data hall increases. The return air/water temperature is the key cooling system operation metric in most cases to determine economization potential. Increased levels of economization result in water and/or power savings, but the magnitude of these savings depends on the type of equipment a facility utilizes and the facility's location.

Generally, for air-cooled IT equipment, there are more potential hours of air-side economizer operation available compared to water-side economizer operation. Air-side economization typically uses direct air cooling which has minimal thermal losses and can achieve air cooling between 85 and 95% of the wet bulb depression. Water-side economization, when using evaporative cooling towers, typically cools water to 70 to 75% of wet bulb depression and uses multiple heat exchangers to transfer heat from the IT equipment to the outside environment. The lower heat transfer efficiency of water-side economization requires lower ambient temperatures to operate and therefore have a lower estimated potential hours for most locations.

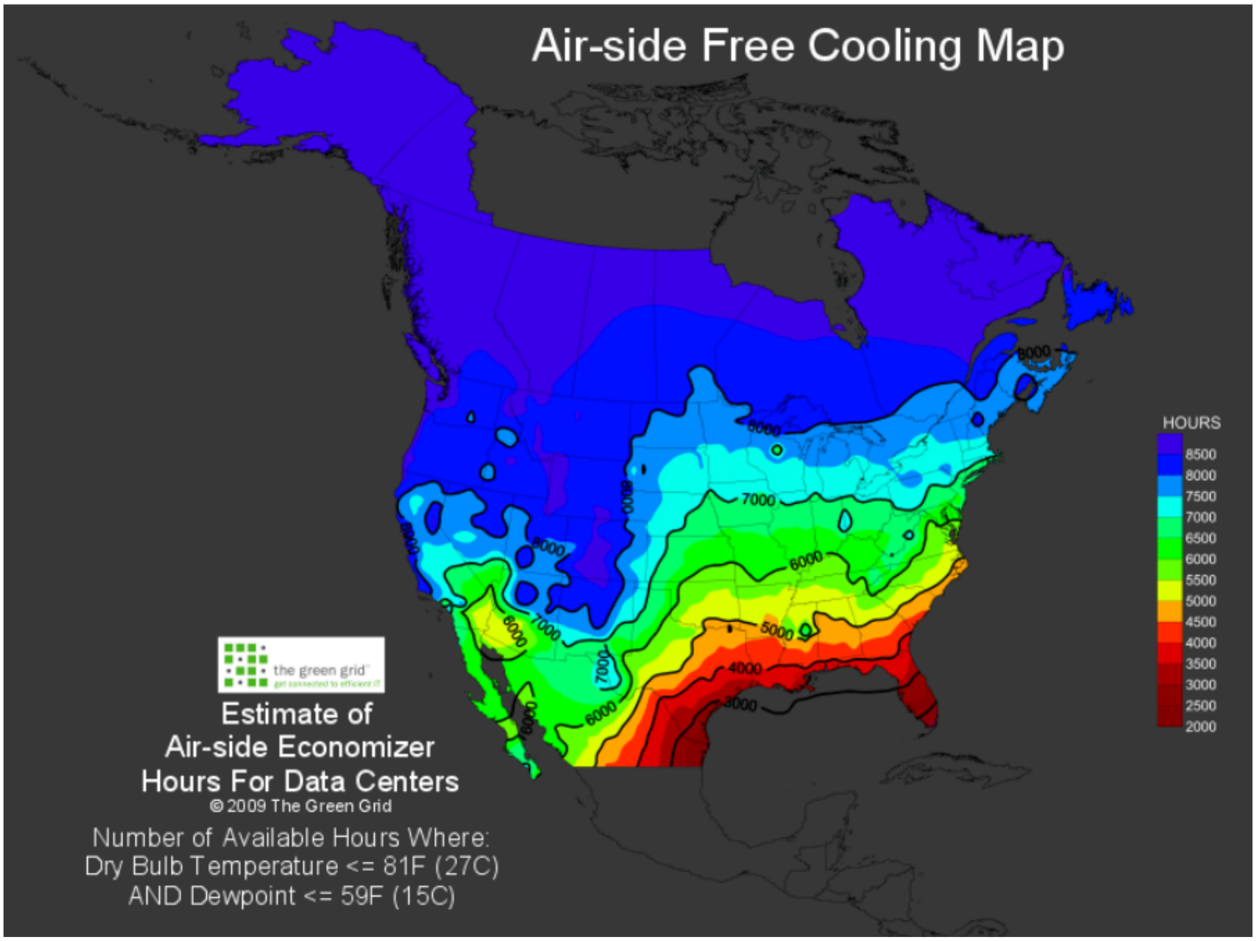

Figure 6. Estimated hours of air-based economization - U.S. [6]

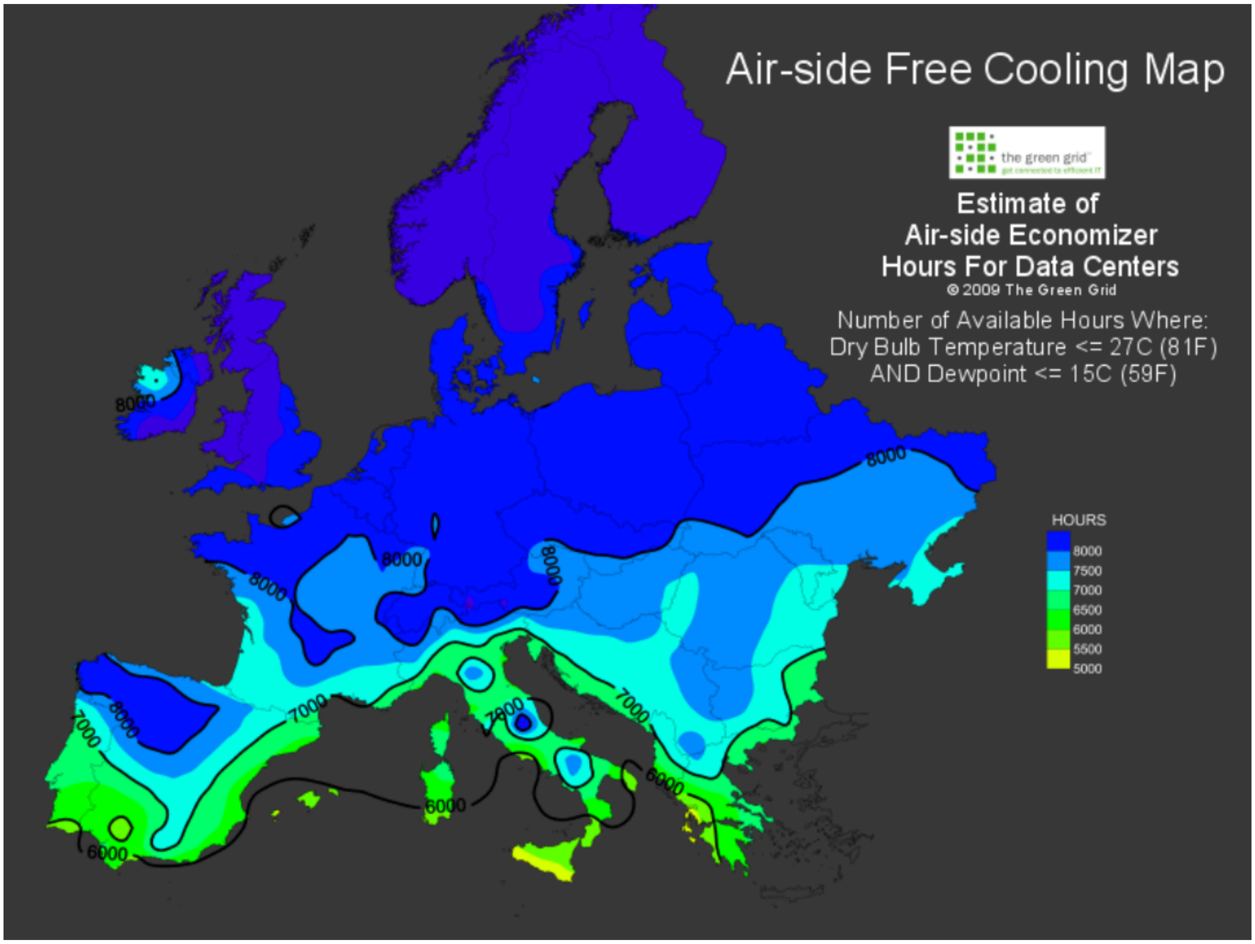

Figure 7. Estimated hours of air-based economization - Europe [6]

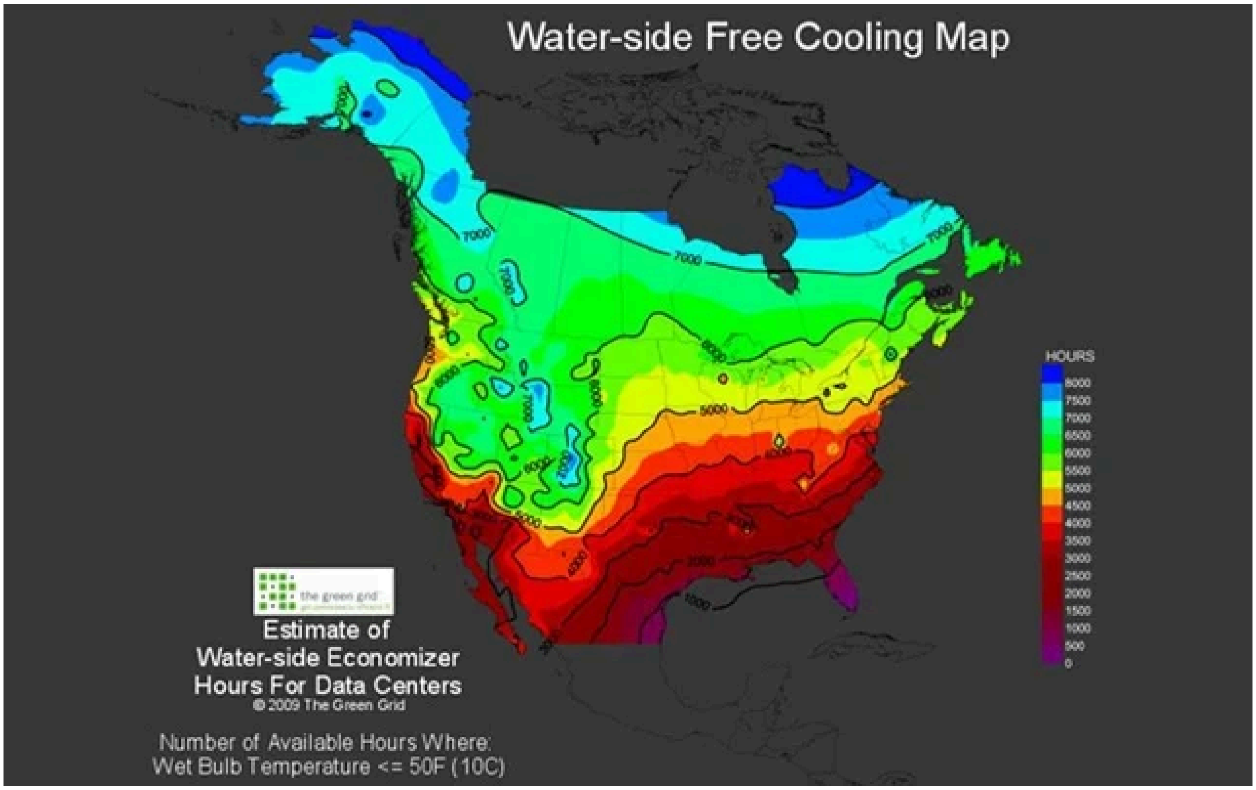

Figure 8. Estimated hours of water-based economization - U.S. [7]

2.1.3 Total Operational Water Consumption

While it was previously stated that evaporative systems use more water but less power, this is only relevant to water consumed directly at the facility. The majority of power generated in the United States requires some level of water consumption, and a common metric for measuring this is the energy-water intensity factor (EWIF). To capture the total operational water footprint of a facility, both the onsite and power generation water consumption should be considered. These two values can be referred to as site and source water consumption, respectively.

The Green Grid has proposed two metrics to account for source and site water consumption. The water usage effectiveness (WUE) is defined as the site water consumption per ITE load and has units of (L/kWh). The total water consumption, which consists of both site and source water consumption, is captured as:

$$WUE_{source} = WUE + EWIF \cdot PUE$$

where the power usage effectiveness (PUE) is the ratio of total facility electricity load divided by ITE load. One can see that the first term in this equation accounts for the site water consumption, whereas the second term represents the source water consumption. The EWIF depends on both geographic region and fuel type, and site planning should consider the evolving trends by utilities in their energy generation portfolio.

Non-evaporative systems use more electricity than evaporative systems, so non-evaporative systems therefore have a higher source water consumption. The total amount of water consumed by non-evaporative systems will be linearly dependent on power consumption, assuming that the EWIF remains constant. Total operational water consumption will peak in the summer when electricity usage is highest.

Evaporative systems will have a more consistent source water consumption throughout the year as the electricity stays consistent. The site water consumption, however, increases during hotter periods as more water is evaporated at the facility to meet cooling demands.

The specific ratio of source water consumption versus onsite water consumption is dependent on the local climate, EWIF for the region, and the efficiency of the equipment used. Facilities located in areas with a high EWIF may see non-evaporative designs with an equal or higher amount of total operational water consumption than evaporative designs when source water consumption is accounted for. However, facilities in areas with a low EWIF or highly efficient non-evaporative cooling equipment may maintain a lower total operational water consumption compared to evaporative systems.

2.1.4 Water Stress

A common approach to examining the environmental capacity is through the water scarcity footprint (WSF), which represents the water footprint multiplied by a water scarcity indicator. The WSF due to operations commonly includes contributions from site water and source water. It is therefore important to consider the water scarcity impacts indirectly associated with the consumed power on site.

Two primary metrics could be used as the water scarcity indicator. The life cycle assessment (LCA) community has recommended the Available WAter REmaining metric, or AWARE [8]. The AWARE factor, A_{CF}, is defined as:

$$A_{CF} = AMD_{ref}/AMD$$

where AMD is the available water minus demand, with a reference value assigned as the weighted average of AMD values for an entire region. The values of A_{CF} are bounded between 0.1 and 100 for water-rich and water-scarce areas, respectively. This metric is the most commonly used indicator in recent academic studies.

A second metric is the Falkenmark indicator [9], which incorporates both natural (climate aridity and the frequency of droughts) and human-made (water consumption and land degradation) factors. Their metric provides a scale of 1 to 5, with 1 indicating minimal water stress, and 5 being "beyond the water barrier," indicating an unsustainable consumption scenario. This indicator is listed in the ISO specification for WUE (ISO/IEC 30134-9).

Per Section 2.1.3, the water scarcity footprint should include elements of on-site water consumption as well as indirect water consumption, such as the water consumption associated with power generation for consumption at the data center. The water scarcity metric for a given stream of water consumption should be associated with the point of use (e.g., on-site or at the power generation source(s)). The most straightforward approach to incorporate the water scarcity footprint associated with grid-based power consumption is the scarce water index (SWI), which incorporates the water scarcity footprint associated with power generation as well as electricity transfers within the grid. The water scarcity usage effectiveness (WSUE) is proposed as the total contributions of source and site water on regional water scarcity footprint divided by IT load. This metric essentially multiplies each point of water consumption by an associated value of A_{CF}. The net result is [10]:

$$WSUE = WUE \cdot A_{CF} + PUE \cdot SWI$$

The Climate Neutral Data Centre Pact (CNDCP) provides guidance on water consumption based on location-specific factors as related to water stress impact. The CNDCP, whose guidance is applied to data centers with initiated construction in 2025 or later, provides water consumption limits based on climate type, water type (potable/fresh, gray, etc.) and stress factor, the latter of which is defined based on the water exploitation index defined by the European Environment Agency Water Exploitation Index for river basin districts (1990-2015).

2.1.5 Facility Location

While many factors impacting a data center's cooling requirements will not be static over the life of the facility the facility's location will not change. The location will have a large effect on the energy and water requirements and their relationship.

For example, a data center located in ASHRAE climate zone A2 (hot and humid) will see limited benefit from evaporative cooling systems. While a data center in climate zone 5B (cold and dry) may greatly benefit from evaporative cooling. This is because evaporative cooling works best in dry regions, as opposed to humid regions. To provide another example, if a data center were to be located in climate zone 6B (cold and dry), while evaporative cooling may work well, air-based cooling should also be considered to take full advantage of the cool ambient conditions.

An additional set of factors to look at, especially when determining water availability is 1) environmental capacity, 2) infrastructural capacity, and 3) water rights. These three factors, or variables, should be examined during the site selection process.

Environmental capacity: a region's ability to consistently provide water. The AWARE factor, Falkenmark indicator, and source water index are all dependent upon geographic location, so location has a strong influence on water scarcity footprint.
Infrastructure capacity: a region's ability to physically deliver water resources from the source to the project location. If you're project is located in or near to a water utility, the utility's capacity to deliver water should be examined. Even if the environmental capacity is high, if the water utility doesn't have, for example, unallocated transmission, distribution, or treatment capacity, then you won't be able to access water without upgrades to the utility's infrastructure.
Water rights: the ability of the facility owner to withdrawal water from a source. If a project sources water from a well or river (instead of a water utility), then water rights likely must be purchased. If you're sourcing water from a utility, then it is advisable to determine how much water rights the utility has vs. its demand.

The table below demonstrates the relationship of these three variables.

Table 2. Cooling Strategies: Operational Considerations

Environmental Capacity	Infrastructural Capacity	Water Rights	Water Available?	Mitigation	Water available after mitigation?
Sufficient	Not Sufficient	Not Sufficient	No	Upgrade infrastructure and purchase water rights	Yes
Sufficient	Sufficient	Not Sufficient	No	Purchase water rights	Yes
Sufficient	Sufficient	Sufficient	Yes	N/A	Yes

As the table illustrates, the environmental capacity is crucial to implementing a water-based cooling system, and mitigation strategies are required if the other categories are not met.

Another variable to consider is water quality, which may affect the amount of water withdrawals required to meet cooling needs. For example, if the water quality of an aquifer is very hard, the cooling systems will either require chemical treatment, mechanical treatment with an associated waste stream, or the system will have to be operated in a less water-efficient manner.

2.2 RELATIONSHIP BETWEEN ENERGY AND CARBON FOOTPRINT

Data centers significantly impact carbon emissions, particularly in terms of energy consumption, primarily from ITE power demands and cooling. Scope 1 emissions may be produced directly from on-site fuel combustion (for example, operating diesel or natural gas backup generators, or alternatively operating a dedicated purpose-built power plant for data center energy supply). Scope 2 and Scope 3 emissions arise from electricity usage from the power grid and upstream/downstream activities like supply chains and end-of-life waste. Understanding the impact of data centers on Scope emissions is an important first step to achieve significant mitigation of carbon footprints. In particular, with the increasing adoption of Large Language Models and Deep Learning Algorithms in Artificial Intelligence, there are distinct differences in carbon intensity between traditional data centers (used primarily for storage) and AI-optimized data centers (focused on high-performance computing, training, and inference), each requiring tailored strategies to minimize emissions.

2.2.1 Traditional Data Centers for Storage

2.2.1.1 Energy Consumption

Traditional storage-oriented data centers typically rely on lower-intensity workloads, which translates into moderate energy consumption. Their cooling requirements are met primarily through air-based ITE cooling, which is less energy-efficient compared to liquid cooling solutions used in AI-optimized centers. Cloud, Colocation and Enterprise Data Centers are often designed to maintain consistent performance without the need for high computational power, with a limited use of compute accelerators such as GPU cards and a much higher proportion of networking and storage. Even so, for large scale cloud and colocation data centers, the sheer size and number of servers required for storage can still result in significant energy use.

2.2.1.2 Carbon Footprint

The carbon footprint of traditional data centers is heavily tied to the energy grid they are connected to. In regions where electricity is generated predominantly from fossil fuels, Scope 2 emissions are significantly higher. Strategies such as signing power purchase agreements (PPAs) with renewable energy providers, or locating these data centers in regions with a cleaner energy mix (e.g., hydroelectric or wind-powered grids), can help mitigate carbon emissions.

2.2.1.3. Water Consumption

Traditional data centers that use evaporative cooling can have a considerable water consumption footprint, particularly in water-scarce regions. Shifting to air-cooled systems, while increasing electricity usage, can reduce water consumption but may raise the overall carbon footprint due to the increased energy demand.

2.2.2 Carbon Mitigation Strategies in Traditional Data Centers

- **On-Site Renewable Energy**: Installing solar panels or wind turbines can directly offset Scope 2 emissions.
- **Energy-Efficient Design**: Maximizing the use of passive cooling techniques (such as air-side economization) can reduce energy consumption and subsequently reduce carbon emissions.
- **Waste Heat Recovery**: Using waste heat from traditional data centers for other processes (such as district heating) reduces overall energy demand and improves sustainability.

- **Hydrogen Fuel Cells**: Hydrogen, especially hydrogen developed using renewable energy resources (e.g., “green” hydrogen created using electrolysis) provides an attractive alternative to traditional diesel generators for backup power. The hydrogen, stored on site, can be used to power fuel cells for backup power when needed, thereby removing the diesel generators and battery banks. Carbon emissions for backup power generation are therefore reduced since diesel combustion is avoided.

2.2.4 AI-Optimized Data Centers for High-Performance Computing (HPC)

2.2.4.1 Energy Consumption

AI-optimized data centers demand significantly more power than traditional storage facilities, given the intensive nature of AI training, inference, and HPC tasks. Increasingly exceeding 100kW of power per rack, these data centers often employ specialized hardware like GPUs and TPUs that consume large amounts of electricity and generate a substantial amount of heat. AI workloads, particularly when training large language models (LLMs) and deep learning algorithms, can exponentially increase power demand, placing a premium on efficient cooling systems.

2.2.4.2 Carbon Footprint

Given their extreme power consumption, AI data centers tend to have a larger carbon footprint unless actively mitigated through carbon offsetting or renewable energy integration. While AI Data Centers are much more energy efficient in terms of number of Floating Point Operations per kWh, the size of the models involved (often exceeding 1 Trillion parameters) results in an increase of 10x to 100x the power consumption of traditional data centers.

Direct-to-chip liquid cooling and immersion cooling systems, while reducing cooling power needs, also increase the infrastructure’s complexity and may have higher embodied carbon due to the materials required for the cooling technology. AI data centers can offset cooling impacts on carbon footprint by locating in regions with strong renewable energy portfolios within the power mix of the grid, hence reducing Scope 2 emissions.

2.2.4.3 Water Consumption

The cooling requirements of AI-optimized data centers often lead to greater water consumption if evaporative cooling techniques are used. However, liquid cooling solutions that operate at higher temperatures can reduce the reliance on water-intensive cooling processes, lowering both water consumption and electricity usage.

2.2.5 Carbon Mitigation Strategies for AI Data Centers

2.2.5.1 Optimized Reference Designs

Innovative reference designs can promote standardization of best practices in AI data centers, and help minimize the carbon footprint while addressing the growing energy demands of AI workloads. These designs provide a blueprint for deploying high-density AI clusters, optimizing power and cooling infrastructure, and improving energy efficiency. A recent example is the collaboration between Schneider Electric and Nvidia that outlines multiple scenarios of power density and cooling optimization in the deployment of AI data centers.

2.2.5.2 Liquid Cooling

Liquid cooling not only improves energy efficiency but also enables the use of waste heat for industrial or residential heating, further decreasing the carbon footprint. However, liquid cooling requires a significant shift in the design and operation of data centers, and may have additional impacts on carbon emissions throughout the supply chain.

2.2.5.3 Strategic Siting

By locating AI data centers in regions where grid electricity is predominantly sourced from renewables, Scope 2 emissions can be drastically reduced. Additionally, positioning these centers near clean energy sources, like hydroelectric power or nuclear, allows for on-site or nearby renewable and low-carbon energy integration.

2.2.5.4 PPAs and Green Tariffs

Securing long-term power purchase agreements for renewable energy directly addresses Scope 2 emissions, reducing the carbon footprint associated with high-performance computing tasks.

2.2.5.5 AI Workload Optimization

Scheduling AI workloads during periods of lower energy grid carbon intensity (e.g., when renewable sources are contributing more power to the grid) can further reduce the carbon footprint.

2.2.6 Conclusion

The impact of data centers on carbon emissions varies significantly depending on the type of facility, its load profile, and the active computing, storage, networking, training and inferencing workloads it handles. Traditional storage-based data centers can mitigate their carbon footprints through renewable energy use and efficient cooling designs, while AI-optimized data centers must prioritize energy-efficient cooling, workload management, and renewable energy integration to reduce their carbon footprint. Regardless of the nature of the data center facility, standardized metrics for reporting of Scope 1, 2, and 3 emissions should be implemented across the industry. In particular, we make the following recommendations:

1. Increases in onsite energy consumption in AI data centers directly lead to a higher carbon footprint, largely due to the additional energy required to operate cooling systems. As AI workloads from high powered GPUs and TPUs generate significant heat, especially in high-density clusters, cooling becomes more energy-intensive, thus increasing the overall carbon emissions of the facility.
2. Different types of cooling equipment have varying inherent carbon footprints based on their energy efficiency and cooling methods. For example, air cooling is less energy-efficient than liquid cooling, which can handle higher densities at lower energy costs. More research is needed to fully understand and quantify these differences to optimize cooling system choices for sustainability.
3. The carbon footprint of data center operations can be significantly reduced through the use of on-site renewable energy sources, power purchase agreements (PPAs) that secure clean energy, by strategically locating the data center in regions with low grid-based emissions, or by replacing diesel generators with fuel cells for backup power. These measures help offset the environmental impact of energy consumption, making data center operations more sustainable.

2.3 RELATIONSHIP BETWEEN WASTE HEAT REUSE AND ENERGY, WATER, AND CARBON

2.3.1 Overview

In a data center where free-cooling is used instead of mechanical cooling (Section 2.1.2), an evident energy saving happens: compressors stop working and only pumps and fans are employed to evacuate the heat to the environment. On top of that, if evaporative cooling is employed, a certain amount of water is consumed for cooling down the return air coming from the white space, usually in an indirect way, allowing to increase free-cooling periods since this approach effectively increases the ambient temperature at which the free-cooling can be utilized. Similarly, any reuse of waste heat to reduce periods of mechanical cooling operation conserves energy on site. Using the excess heat of the data center implies two main scenarios:

1. The waste heat is circulated out of the IT space without modifying its characteristics (same temperature, pressure, state…)
2. The waste heat suffers a physical modification in its way out from the IT space

2.3.2 First Scenario: No Modification of Waste Heat Characteristics

In this scenario, the usage of the excess heat allows:

- Less mechanical cooling usage at the data center. This implies:
    - A direct **lower energy usage** at the compressor and the fans, hence indirectly reducing the carbon footprint and water usage associated with the energy supply.[1]
    - A direct **lower usage ratio** of the mechanical equipment, potentially extending its life as well as the refrigerant. This results in (1) reducing the embodied carbon and water footprints associated with the manufacturing of equipment replacements, and (2) reducing the potential direct carbon footprint associated with refrigerant leaks.
- Less evaporative cooling usage at the data center. This implies:
    - A direct **lower energy usage** for the evaporative system (e.g., pumps and sprays), thereby indirectly reducing the carbon footprint and water usage associated with the energy supply.
    - A direct **lower usage ratio** of evaporative cooling equipment, thus extending its life. This results in reduced embodied carbon and water footprints associated with the manufacture of equipment replacements.
    - A direct **lower water usage** directly associated with the evaporation of the water.
- Less thermal energy at the heat offtaker. This implies:
    - A direct **lower energy usage** for the heating equipment (e.g., boiler, heat pump, and solar thermal equipment). This reduced energy consumption correspondingly lowers the carbon and water footprints associated with their energy supply.
    - A direct **lower usage ratio** of the heating equipment, resulting in a potentially longer equipment life. This result indirectly lowers embodied carbon and water footprints associated with the manufacture of equipment replacements. Furthermore, this result reduces the potential direct carbon footprint associated with refrigerant leaks in the case of refrigerant-based thermal machines.
    - A direct **lower water usage** directly associated with the evaporation of the water occurring at the heat generation, if applicable.
    - A **potential higher water usage** in case of leaks due to a long interconnection between systems.
    - A **potential water gain** if the heat is used for dirty/salty water treatment.
    - A direct **lower carbon footprint** directly associated with the burning of fuels for heating production.

[1] Waste heat reuse will generally imply the usage of a liquid-to-liquid heat exchanger. This equipment might add more resistance to the cooling circuit as the cooling tower/dry cooler. This aspect should be carefully considered for accurate energy balance comparisons.

2.3.3 Second Scenario: Modification of Waste Heat Characteristics

In the second scenario, where the physical characteristics of waste heat are modified, the reuse process introduces additional layers of complexity, especially when considering the total energy balance between the data center and the heat offtaker as a global project. Typically, the offtaker would manage any modifications to the waste heat, such as increasing temperature, pressure, or changing its state, since they are generally more specialized in these transformations. However, situations where legal requirements mandate heat reuse, or where the resulting heat has higher market value, might incentivize the data center to invest in and operate the necessary modification devices. We have then two possible subscenarios when employing thermal machines or other devices dedicated to increase the temperature, pressure, change of state and/or transforming the waste heat into a different product[2]:

2.3.3.1 Subscenario 1: Less Energy Required.

If the modification of the waste heat (e.g., through heat pumps) requires less energy than alternative approaches, this results in a direct energy savings from a global perspective. Even though additional machinery is employed, overall energy consumption, carbon emissions, and water usage are reduced. This approach might increase the usage ratio of the machinery involved, which could raise embodied carbon and water footprints due to the need for more frequent replacements. However, recent advances in **energy-efficient heat pumps and integrated thermal energy systems** have shown that these modifications can lead to significant net reductions in energy use, particularly when compared to the traditional use of chillers and cooling towers at the data center combined with boilers at the offtaker. For instance, heat pumps in data centers operating in cold climates like Finland have successfully captured low-grade waste heat and transformed it into usable high-grade heat for district heating networks, significantly cutting energy demand.

2.3.3.2 Subscenario 2: Equal or More Energy Required.

In cases where more energy is required to reuse waste heat than the offtaker would have used independently, there must be an economic, ecological, or regulatory benefit to justify the project. This might occur when using inefficient transformation technologies or when additional heating equipment is necessary. An example is when high-temperature heat reuse is mandated by local regulations, as seen in urban areas of Denmark, where district heating systems require higher temperatures than those typically generated by data center waste heat. If the transformation process consumes equal or more energy than simply producing heat independently, the justification for the system often lies in reduced carbon emissions or alignment with sustainability goals.

2.3.3.3 Recent Developments Impacting Waste Heat Reuse

The rise of innovative waste heat utilization technologies has broadened the potential for data centers to participate in circular energy economies. For instance, new research shows that **waste heat recovery systems integrated with data centers can be optimized using AI** to balance energy loads and predict heat demands, further reducing the overall carbon footprint. These systems can also support nearby industries or agricultural operations, such as greenhouse heating, where the carbon and water benefits are amplified by aligning waste heat supply with a constant demand. An example of this is Microsoft's data center in the Netherlands, which

[2] Waste heat can be transformed into other energy carriers such as cooling energy (using absorption or adsorption cooling) or electricity (using Organic Rankine Cycles or other isothermal cycles)

utilizes waste heat to warm local greenhouses, thereby turning the data center into a hub of energy efficiency and environmental benefit.

Waste heat reuse, when not requiring additional energy, is one of the most sustainable methods of reducing the data center's environmental impact. When additional machinery is employed, careful balancing of energy efficiency and market value becomes critical to ensuring that the reuse process remains environmentally and economically viable.

2.3.4 Use of Waste Heat for Water Reclamation

Much of the water losses in data centers occur through evaporation-aided cooling (Section 2.1.1), where water is absorbed into an airstream, thereby increasing the airstream's humidity ratio. Such examples include the outlet airstream of a cooling tower and the internal airstream from direct air cooled data centers. It is possible to reclaim some of the lost water in this humid airstream using condensation-based recovery. The strategy for condensation-based recovery can include the use of waste heat recovery in three ways:

1. The waste heat is used to generate chilled water using an absorption or adsorption chiller system. The chilled water temperature is below the humid airstream's dewpoint temperature to enable external condensation. The chilled water stream is cooled using the evaporator in the absorption/adsorption chiller system, and a cooling coil is placed in the humid airstream.
2. The waste heat is used to generate electricity using an organic Rankine cycle (ORC). The electricity is then used as an input into a thermoelectric (Peltier) cooler, where the cold side of the cooler falls below the humid airstream's dewpoint temperature. Note that ORCs require a significant temperature difference (at least 40 C) across the expander to function properly, which may therefore cause issues in operation during warm weather unless supplemental energy is applied (Section 2.3.3.2).
3. The waste heat is used to generate electricity using an organic Rankine cycle. The electricity is then used as an input to an electrostatic mesh placed over a water droplet plume such as that leaving a cooling tower. This approach has a strong potential for water reclamation but is most viable in humid climates where water droplet plumes are most frequent.

These approaches generally involve the transformation of waste heat following Section 2.3.3 to ensure a high-enough waste heat temperature to provide sufficient thermal power into the evaporator of the absorption chiller or organic Rankine cycle, or the desorber of an adsorption chiller. It should be noted that the required transformation has an embedded water footprint following Section 2.1.3, so care must be taken that employing the transformation does not result in a higher Scope 1 and 2 water footprint when considering the addition of virtual water footprint minus the reclaimed water. Finally, the use of reclaimed water may require treatment before reintegration into a cooling system such as the cooling tower spray or evaporative cooler media following Section 2.3.2.

2.3.5 Use of Waste Heat for Carbon Recovery

The use of waste heat could also potentially enable the facility to reclaim some of the carbon footprint associated with power consumption. This goal is achieved in several ways that can be broken down into categories of natural methods and direct air capture.

2.3.5.1 Natural Methods

These methods involve the use of natural materials or organisms. Such strategies include:

1. Use of waste heat to heat a greenhouse containing seedling plants. These plants absorb CO2 and can be either planted adjacent to the data center. One approach to optimize CO2 capture is Miyawaki afforestation [11] to maximize carbon removal density through strategic planting in minimal spaces. Alternatively, regenerative agriculture could be used to restore the carbon in the soil.
2. Apply method #1 to grow edible produce for sale to the surrounding community. This is especially effective in food desert areas such as urban neighborhoods or arid rural communities.
3. Proliferate microorganisms to absorb greenhouse gas emissions. Many microorganisms absorb CO2 to survive. Extremophiles are a specific class of microorganisms that absorb higher GWP gases such as methane.
4. Create natural alternatives to synthetic materials. Past examples of this include the intermixing of natural materials in concrete, thereby reducing the per-volume carbon emissions associated with the manufacturing of concrete walls. Likewise, natural materials could be introduced as alternatives to steel to avoid the emissions associated with manufacturing of steel.

2.3.5.2 Direct Air Capture

Direct air carbon capture and sequestration (DACCS) is a method to remove the carbon dioxide embedded in the ambient airstream surrounding the data center, followed by sequestration of the carbon to prevent re-release into the atmosphere. DACCS involves the flow of the airstream over a bed of material that absorbs CO2, after which the material is placed in permanent storage. This method is most effective near airstreams with high CO2 content such as that exiting a combustion-based power plant.

2.3.6 Renewables, Waste Heat Recovery, and Large Data Centers

Within the circular economy, the use of renewables and the reclamation of industrial waste heat are important concepts. This is especially true for the energy-heavy datacenter industry, since if the waste heat can be used by other industrial or residential entities, there will be less strain on the local power grid as well as on the environment. Some considerations are:

- Where renewable energy is abundant and steadily produced, there is more flexibility for future expansion of the data center, and the energy price will be less fluctuating.
- Obtaining permissions for new data centers today is often difficult, because of the strain on the local power grid, and the relatively few work opportunities a data center brings on a long-term basis. From a circular economy (or "industrial symbiosis") point of view, it is therefore easier to motivate the establishment of a large data center to local officials, if there is a demand for the waste heat.

2.4. SUMMARY TABLES OF INTERPLAY

Tables 3a-3c below contain trends drawn from group consensus, physical insights, and experience, for cooling strategies in comparison to a base case featuring only CRAC-based cooling with no heat recovery, with heat discarded to ambient air. The symbols are major decrease (D), minor decrease (d), no change (-), minor increase (u), major increase (U), and uncertain (?). Note that general trends are shown, but there may be cases that go against the stated trends. Also, the magnitudes provided are general in nature and do not represent all cases. These tables, especially those marked uncertain, help identify areas for further investigation to provide additional guidance from the workstream in future white papers. Water stress impact can furthermore be considered for future work in conjunction with the trends identified in the tables. The viability of certain strategies, for example, depend on the local environment. Note also that some strategies are strongly encouraged by the industry: ASHRAE encourages economization, and there is a European requirement for heat reuse in data centers. Furthermore, liquid cooling is becoming necessary to handle high rack heat loads (> 100 kW/rack), so the sustainability benefits, albeit a secondary motivation, are nevertheless welcome.

Note that Table 3b does not include one strategy: net water balance. This approach applies community-based water conservation projects to offset data center water consumption. This strategy is excluded in the table since the columns allude to data center-specific consumption. The impact of these programs does not alter the data center energy, water, and carbon characteristics specific to the data center itself, but it provides a means to holistically (data center + surrounding community) reduce water consumption. The impact of these programs on holistic energy use and carbon footprint requires investigation and is beyond the scope of this paper.

Table 3a. Cooling Strategies: Operational Considerations

Strategy	Energy	Water (on-site, upstream, total)	Carbon (operational, inherent, total)	Notes
Evaporative cooling	D	U, D, ?	D, ?, ?	Adiabatic cooling reduces energy but adds on-site water
Liquid cooling - cold plates	d	-, d, d	-, d, d	Assuming no use of wet cooling towers. Could decrease water since less adiabatic cooling due to higher water temps
Liquid cooling - immersion	d	-, d, d	-, d, d	Assuming no use of wet cooling towers. Could decrease water since less adiabatic cooling due to higher water temps
Operational strategies/decisions	D	?, ?, ?	D, ?, ?	Assume goal is to minimize energy consumption
Airside economization without evaporative cooling	D	-, D, D	-, D, D	
PPAs	-	-, -, -	D, -, D	
Hydrogen (Fuel Cells)	-	d, -, d	D, -, D	Replaces diesel generators, on-site water generation

Nuclear (SMRs)	-	?, ?, ?	D, ?, ?	Water aspects depend on cooling system type and need further investigation
KEY: major decrease (D), minor decrease (d), negligible change (-), minor increase (u), major increase (U), uncertain (?)				

Date: March, 2026

Table 3b. Cooling Strategies: Geographical Considerations

Strategy	Energy	Water (on-site, upstream, total)	Carbon (operational, inherent, total)	Notes
Geographical location to minimize operational energy use	D	?, ?, ?	?, ?, ?	Site with favorable climate may also have large Scope 2 carbon emissions and water consumption
Geographical location to minimize operational carbon footprint	?	?, ?, ?	D, -, d	External air temperature/humidity levels dictate energy and on-site water consumption
Geographical location to minimize water scarcity impact	?	?, ?, D	?, -, ?	
Onsite renewables	-	-, ?, ?	D, -, D	On-site solar is well-developed. On-site wind is less developed, deployed at a smaller scale, and is less financially viable
KEY: major decrease (D), minor decrease (d), negligible change (-), minor increase (u), major increase (U), uncertain (?)				

Table 3c. Cooling Strategies: Waste Heat Recovery Considerations

Strategy	Energy	Water (on-site, upstream, total)	Carbon (operational, inherent, total)	Notes
Waste heat recovery for energy reuse	D	?, d,d	D, u, d	
Waste heat recovery for carbon removal	-	-, -, -	D, -, D	Ideal scenario (no added energy). DC waste heat would likely need to be boosted for carbon removal, but AI-based hardware may get hot enough.
Waste heat recovery for water reclamation	-	D, -, D	-, -, -	Ideal scenario (no added energy)
Waste heat recovery for biodigesters	-	?, ?, ?	D, u, D	Preheating biodigesters can be performed year-round
KEY: major decrease (D), minor decrease (d), negligible change (-), minor increase (u), major increase (U), uncertain (?)				

Table 3d. Non-Cooling Strategy Considerations

Strategy	Energy	Water (on-site, upstream, total)	Carbon (operational, inherent, total)	Notes
Improved power distribution efficiency (e.g., DC-powered data centers)	d	-, d,d	d, ?, d	
KEY: major decrease (D), minor decrease (d), negligible change (-), minor increase (u), major increase (U), uncertain (?)				

3. Examples to Illustrate Interplay

Hypothetical scenarios are now provided (one for each of the tables in Section 2.4 to illustrate how decisions affect each of energy, water, and carbon.

3.1 EXAMPLE: COOLING STRATEGIES - OPERATIONAL CONSIDERATIONS

Two 1 MW data centers are located in Phoenix, AZ:

- Data Center A employs waterless cooling via CRAC units (base case). The data center contains a PUE of 1.6, so the cooling power consumption is 600 kW, or 5.3 million kWh/yr. The EWIF associated with Phoenix, AZ is 8.3 L/kWh, so the annual Scope 1 & 2 water consumption by the data center is 120 million L/yr.
- Data Center B employs evaporative cooling, which reduces the PUE to 1.2 but contains a WUE of 1.8 L/kWh. This reduces the cooling power consumption to 200 kW, or 1.8 million kWh/yr. The virtual water consumption is now 88 million L/yr, and the on-site water consumption is 16 million L/yr, so the total Scope 1 and 2 water consumption is 104 million L/yr, or a reduction of 16 million L/yr compared to the base case. Furthermore, the carbon emission factor is 0.26 kg CO_2/kWh, so the emissions are reduced from the base case value of 3500 metric tons CO_2/yr to 2700 metric tons/yr for the evaporatively cooled data center.

3.2 EXAMPLE: COOLING STRATEGIES - GEOGRAPHICAL CONSIDERATIONS

A company seeks to build a 1 MW CRAC (waterless) cooled data center and is considering locations in Florida or Colorado. The cooler climate associated with Colorado reduces PUE in two ways: first, the lower ambient temperature raises the coefficient of performance of the CRAC units; and second, the cooler climate enables more hours of airside economization. If airside economization is used with evaporative cooling when the ambient conditions fall in the ASHRAE Recommended range, then the PUE for the Florida data center is 1.79, whereas the PUE for the Colorado location is 1.61, a power savings of 180 kW or 1.6 million kWh/yr, as shown in Table 4. However, the different effective carbon emission factors for the two locations leads to increased carbon emissions for Colorado of 2,000 metric tons per year compared to the Florida location. The water scarcity footprint for Colorado is also much higher due to the relative lack of available water in that state. The same analysis can be applied to compare the data center in any state using compiled data [12].

This example illustrates how geographic factors play an important role in comparing two different geographic locations:

- One location (Colorado) provides more hours of free cooling via airside and water-side economization.
- The use of evaporative cooling may be more effective in one location (Colorado) due to a lower average ambient relative humidity, promoting more evaporation, which leads to energy savings.
- The use of evaporative cooling, especially in conjunction with increased airside economization, leads to more onsite water consumption (Colorado).
- The reduction of onsite energy consumption due to increased airside economization potential (Colorado) leads to reduced carbon emissions.
- The water scarcity impact associated with that increased water consumption, yet reduced energy consumption, depends on the scenario. he increased use of evaporative cooling:
    - Reduces energy consumption and therefore Scope 2 water scarcity footprint.
    - Increases onsite water consumption and therefore Scope 1 water scarcity footprint.

  Therefore, additional work needs to be explored to ascertain which of these competing factors most affects the net change in water scarcity footprint. In this case, the higher water scarcity factor for Colorado compared to Florida results in a net increase in water scarcity footprint. This finding suggests that in general the water scarcity footprint will be higher in water-scarce regions but needs further investigation.

Table 4. Comparison of Two Geographic Locations

Location	PUE	Annual Energy Consumption, million kWh	Annual Water Scarcity Footprint, million L/yr	Annual Carbon Emissions, metric tons/yr
Florida	1.79	15.7	30	12,400
Colorado	1.61	14.1	1,300	14,400
Difference	+0.18	+1.6	-1,270	-2,000

3.3 EXAMPLE: COOLING STRATEGIES - WASTE HEAT RECOVERY CONSIDERATIONS

A 1 MW data center in San Francisco, CA is cooled using two-phase refrigerant cold plates. The refrigerant return temperature is 38 C and is boosted using vapor recompression to 125 C and then run through a LiBr absorption chiller. The process requires 270 kW of the compressor and produces chilled water with a cooling capability of 510 kW for external loads [13]. This result means that the cooling system requires 270 kW, yet the net effect of the cooling system on the data center and surrounding buildings is a reduction of 240 kW of cooling, which equates to 80 kW of electricity input for a cooling system with a COP of 3.0. The saved 80 kW of energy is equivalent to 6 million L/yr of water footprint and 160 metric tons per year of carbon emissions.

3.4 EXAMPLE: OTHER CONSIDERATIONS

A data center with an IT load of 300 kW in San Francisco can be designed with either a traditional alternating current (AC) power infrastructure or an alternative direct current (DC) power infrastructure. The AC data center has a total PUE of 2.16, whereas the DC data center has a PUE of 1.89 due to the reduced number of power transitions in the DC-based power distribution network [14]. The energy savings when using the DC-based system are 700,000 kWh/yr. The associated water and carbon savings are 6 million L/yr and 160 metric tons per year, respectively.

4. Compliance with Open Compute Project Tenets

OPENNESS

The information contained in this document is intended for public dissemination. Values used in example calculations are illustrative in nature and not taken to represent measured data in real situations.

EFFICIENCY

The concepts brought about in this document illustrate how improvements in energy efficiency affect environmental burden.

IMPACT

The calculations discussed in this document illustrate how data center environmental burden can be mitigated through cooling system design decisions.

SCALE

The concepts can be applied to any type of data center provided that operational data and location-specific environmental metrics are known.

SUSTAINABILITY

Reducing environmental burden helps to strengthen the sustainability of data centers.

5. Conclusion

The conclusions gleaned from this white paper are:

- Care must be taken when choosing various strategies to improve energy efficiency, reduce carbon footprint, or reduce water footprint. Many strategies improve one area while negatively affecting others.
- Numerous strategies for reducing environmental burden exist, and all should be considered when examining strategies for designing and operating data centers.

Date: March, 2026

- A holistic view is needed, both in terms of Scope 1 and 2 impacts (i.e., water footprint) as well as in the influence of the data center on the surrounding areas (e.g., through waste heat recovery) in order to obtain an accurate view of the influence of strategies on the various environmental metrics.
- Much work needs to be done to determine the nuances in how various strategies influence one or more of energy consumption, carbon emissions, and total water footprint.

6. Glossary

AC: alternating current.

A_{CF}: Available WAter REmaining (AWARE) factor.

AMD: availability minus demand.

AI: artificial intelligence.

CNDCP: Climate Neutral Data Centre Pact.

CRAC: computer room air conditioner.

CRAH: computer room air handler.

DC: direct current.

Direct air carbon capture and sequestration (DACCS): a method to remove and store CO2 from an airstream.

DLC: direct liquid cooling.

DX: direct expansion.

EWIF: energy water intensity factor.

HPC: high performance computing.

LCA: life cycle assessment.

LLM: large language model.

ITE: information technology equipment.

Miyawaki afforestation: a method to produce dense forests for carbon removal.

ORC: organic Rankine cycle.

PPA: power purchase agreement.

PUE: power usage effectiveness.

Scope 1: on-site footprint.

Scope 2: footprint associated with upstream production (e.g., power generation).

Scope 3: footprint associated with construction/manufacturing/materials.

SMR: small modular nuclear reactors.

SWI: scarce water index.

TCS: technology cooling system.

WUE: water usage effectiveness.

WUE_{source}: Scope 1 and 2 water usage effectiveness.

WSUE: water scarcity usage effectiveness.

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